CHAPTER 03 — Artificial Intelligence Foundations

Ethics in Artificial Intelligence

Guiding automated decisions, eliminating algorithmic bias, and safeguarding human values.

Focus Area	Target
Responsible Tech	**Fairness & Privacy**

CORE CONCEPTS

Understanding Ethics and Morals

Ethics

Societal rules and standards guiding what is fair, just, and acceptable across communities.

External systemic framework

Morals

Personal beliefs shaped by upbringing and culture, directing individual conscience.

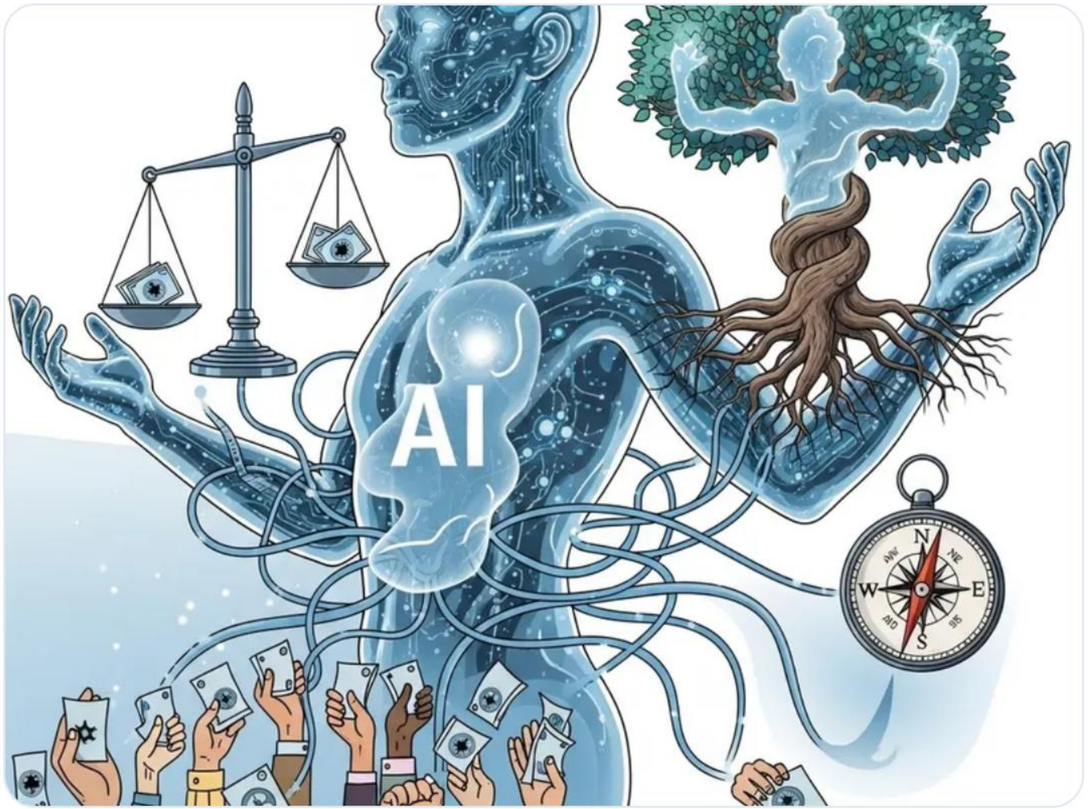
Internal personal conviction

AUTONOMOUS AGENCY

Why AI Systems Require Ethics

AI models learn patterns to make human-like decisions across medical diagnosis, driving, and finance.

- **High Impact:** Algorithms drive critical life outcomes.
- **Autonomous Errors:** Flawed models propagate errors rapidly.
- **Value Alignment:** Systems must uphold societal justice.



GUIDING FRAMEWORK

Six Core Demands of AI Ethics

Fair
Equal treatment without favoritism.

Private
Protected data confidentiality.

Transparent
Clear rationale for automated outputs.

Accountable
Definitive responsibility when systems fail.

Safe
Proactive prevention of societal harm.

Unbiased
Free from discriminatory assumptions.

FUNDAMENTAL AXIOM

”Biased Data will create Bias Decisions.”

When flawed assumptions enter the training set, algorithms reinforce and amplify discrimination.

Input Flaws
Skewed historical records

Output Disparity
Unfair real-world penalties

CASE EVIDENCE

Real-World Manifestations of AI Bias

01

Recruitment Bias

Amazon discontinued an automated hiring tool in 2018 after discovering systematic penalties against female candidates.

Cause: Male-dominated past resumes

02

Healthcare Access

A US healthcare algorithm systematically allocated extra medical resources away from Black patients toward white patients.

Cause: Commercial spending metrics

03

Surveillance Profiling

Security models evaluating potential crime based on demographic traits rather than observable individual actions.

Cause: Skewed arrest records

ROOT CAUSES

How Bias Enters AI Systems

Human Bias

Developers unknowingly embedding personal prejudices into algorithmic weighting.

Flawed Datasets

Under-representation of marginalized groups during data collection.

False Assumptions

Incorrect correlations made about real-world societal behavior.

Missing Audits

Deploying models without rigorous pre-launch bias mitigation testing.

DIGITAL EQUITY

Addressing the AI Access Gap

The Current Reality

AI benefits concentrate among well-funded institutions and high-income classes who afford cutting-edge devices.

Risk: Widening existing economic disparity

The Ethical Mandate

Governments and institutions must provide open infrastructure, enabling equal technological access for all students.

Solution: Democratic digital infrastructure

CORE GUIDELINES

Four Pillars of Responsible Systems

Human Rights

Avoid intrusive surveillance and unlawful discrimination.

Data Privacy

Ethical collection with active consent and robust security.

Inclusion

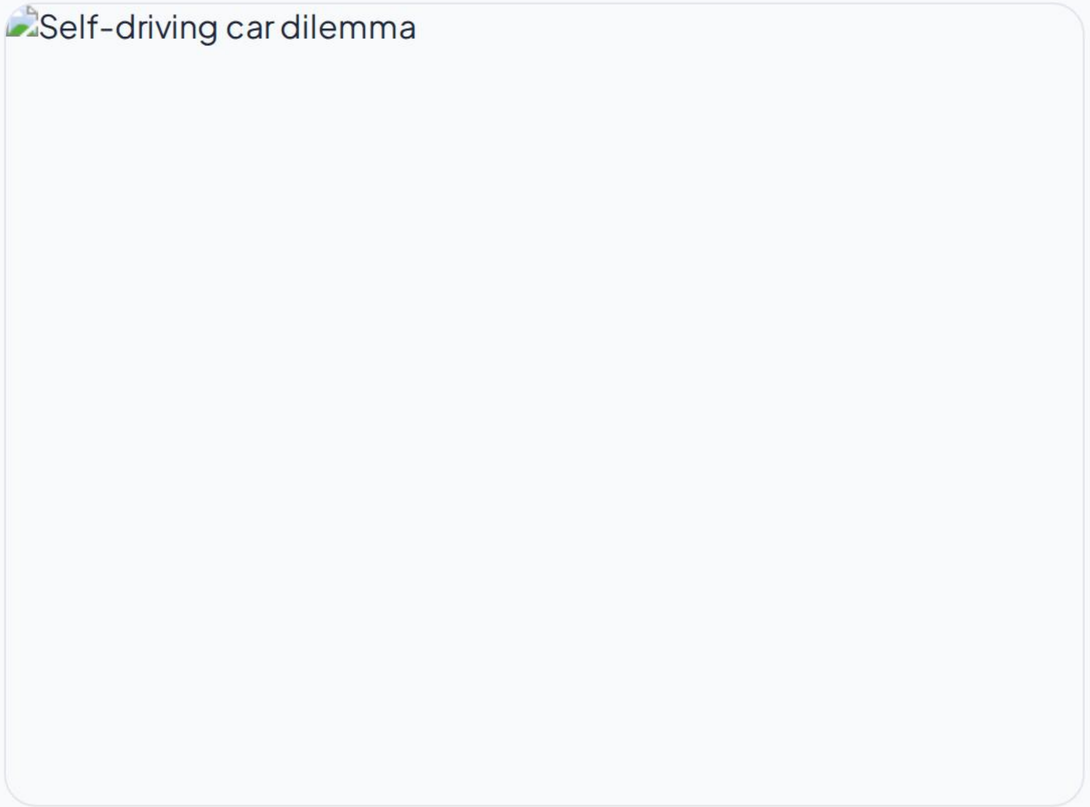
Involve diverse teams to reflect broad user demographics.

Mitigation

Continuous testing to eradicate loan and hiring disparities.

INTERACTIVE INQUIRY

Case Dilemma: Autonomous Vehicles



The Moral Machine Experiment

When an unavoidable crash occurs, how should an autonomous car decide between saving passengers or pedestrians?

Key Question: Who is legally and morally accountable?

Debate Focus: Can human ethics be encoded into pure code?

TERMINOLOGY

Essential Chapter Glossary

Concept	Definition & Scope
AI Ethics	Moral rules governing deployment to ensure fair, safe, and respectful systems.
AI Bias	Unfair or prejudiced outcomes stemming from flawed training data or assumptions.
Training Data	Labelled and annotated datasets used to teach machines how to execute tasks.

LOOKING FORWARD

Building Trust in Technology

Ethical artificial intelligence is not merely a technical safeguard—it is a societal commitment to human dignity.

Open Questions & Discussion

Image Sources



https://media.easy-peasy.ai/03161798-d57c-48ca-b8bf-1ca22575c0ef/75c89fb7-50e9-4d92-ae3e-ea001673b34a_medium.webp

Source: easy-peasy.ai



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